FORTRAN'S ENDURING RELEVANCE IN SCIENTIFIC COMPUTING AND AI INTEGRATION

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Fortran (FORmula TRANslation) lauded by many as the first programming language standard was developed in 1957 by John Backus of IBM. By enabling software developers to use a compiler for each machine vs hand coding each machine, John Backus is considered the “father of Fortran, the first widely used, high-level programming language that helped open the door to modern computing.” (*IBM*, n.d.)

Since its invention, many languages have come and gone. While its syntax and structure are quite archaic in terms of modern programming standards, it is still very much in use in applications requiring rapid scientific and numerical calculations. It is due to its raw simplicity and speed that make it excel at one such application: weather system modeling. The National Weather Service and NASA’s weather monitoring system are great examples of this. ((*GMAO - Global Modeling and Assimilation Office Research Site*, n.d.)

NASA relies heavily on predictive modeling on complex weather systems to determine the optimal time for launching of spacecraft and satellites. Fortran not only provides such a solution, but due to its minimal overhead and well-established codebase, it is no wonder these government agencies do not migrate to more modern solutions. Fortran, being a compiled language, does not require the use of an interpreter and is directly in communication with the hardware that runs it. Afterall a compiled language is encoded in machine language.

Additionally, due to the fact that it is a procedural language instead of an object oriented language, it is significantly easier to trace code execution which avoids the inherent complexities

of object hierarchies. This also makes the memory management of running software significantly more predictable.

While over the years Fortran has adopted aspects of object-oriented programming, in terms of its foundation it remains procedural. This limits some of the advantages seen in modern programming, particularly around code reuse. Classes, inheritance and modularity are the cornerstones of object-oriented programming and allow for faster, more scalable development. The lack of these capabilities in traditional Fortran applications poses significant challenges, especially since fewer developers are proficient in the language. (GeeksforGeeks, 2022)

Migrating this code also allows the use of GPU powered AI models to be leveraged for potentially more predictive models. By leveraging AI, developers stand to see significantly more accurate predictions vs Fortran's modelling limitations, given the right data sets as training data. An example of this is a google product that leverages google DeepMind called GraphCast. (*GraphCast: AI Model for Faster and More Accurate Global Weather Forecasting*, 2025)While this has been demonstrated as possible using Fortran, the interfacing and actual execution is significantly more obscure, and you still run into the clear lack of expertise on the language.

Ultimately, Fortran's predictable memory footprint, compiled efficiency and decades-mature libraries keep it indispensable in areas where mission-critical scientific calculations need to occur. Nonetheless, due to its smaller talent pool, procedural nature and limited naïve support for AI based modeling, migration to other solutions may end up being the best bet. A pragmatic approach would be one of a hybrid system that utilizes Fortran's strengths while downplaying its

weaknesses by retaining proven Fortran kernels and wrapping or replacing components with more advanced AI modules. This would boost performance and reliability to legacy code, while unlocking the flexibility that true object oriented and machine learning solutions can provide.

References

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